INCH-POUND

MIL-PRF-8565/18
w/AMENDMENT 2
11 April 2022
SUPERSEDING
MIL-PRF-8565/18(AS)
w/AMENDMENT 1
13 December 2013

PERFORMANCE SPECIFICATION SHEET

BATTERY, STORAGE, AIRCRAFT, MAINTENANCE-FREE, 24-VOLT, 10-AMPERE-HOUR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-8565.

REQUIREMENTS:

1. Part or identifying number. The part or identifying number (PIN) of the batteries covered by this specification sheet shall be D8565/18-1.

2. Dimensions and weight. The dimensions of the battery shall be as shown on figure 1. The weight of the battery shall be not greater than 33.0 pounds (15.0 kilograms) (see 3.8.3 and 4.5.3 of MIL-PRF-8565). In addition, the weight of any production battery shall vary by not greater than ± 2.2 lb. from the average weight of the three approved qualification sample batteries but may not exceed the 33.0-pound limit.

3. Capacity. The rated capacity of the battery shall be not less than 10.0 ampere-hours (Ah), based on its ability to conform to the life test final discharge. The life test final discharge is a 1-hour discharge at 24 °C ambient temperature with a 10-ampere load (1 C-rate) to a cutoff voltage of 18.0 volts (10 Amp/1.0 hr/24 °C/18.0V). See 3.8.6 and 6.10 of MIL-PRF-8565.

4. Service life. The battery's service life (useful life) in aircraft shall be not less than 3 years.

5. Qualification inspection and conformance inspection. Table I lists the required inspections.

6. MIL-PRF-8565 variance. The battery shall comply with MIL-PRF-8565 except as follows.

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6.1 The following paragraphs are not applicable:

3.6.5 Marking for batteries with heaters.
3.8.7 Constant voltage discharge.
3.8.16 Output performance.
3.8.17 Cold and hot temperature charge.
4.5.8 Constant voltage discharge.
4.5.28 Output performance test.
4.5.29 Cold and hot temperature charge test.

6.2 Modify the following paragraph:

3.5.5 Receptacles. Insert the following after the first sentence: "For D8565/18-1, the female receptacle (front side) shall be dimensionally equal to MIL-PRF-18148/3, style 2. The receptacle may be integrally molded in the container."

6.3 Add the following paragraph:

"3.5.11 Mounting holes. The mounting surfaces that surround the two mounting holes shall be free of any coating for a distance not less than 12.7 mm diameter from the centers of the holes. This shall apply to both sides of each mounting surface (see figure 1)."

6.4 Modify the following paragraph:

3.8.5 Vent tubes. Delete the title and text and substitute: "Carrying strap dimensions and strength. The battery cover shall have two flexible carrying straps located as shown on figure 1 of MIL-PRF-8565/18. The straps shall be a dark color and shall be free of any coating. Adhesive methods shall not be used to attach the straps' mounting supports to the battery. Each of the battery's carrying straps shall be not less than 1 inch (nominal) wide. The straps shall not show any evidence of breaking away from the container or other damage after being tested in accordance with 4.5.5, as modified by MIL-PRF-8565/18. The straps and battery shall also meet the requirements of 3.7 of MIL-PRF-8565."

VERIFICATION

1. MIL-PRF-8565 variance. The battery shall comply with MIL-PRF-8565, except as follows:

1.1 Delete paragraph 4.5.5 in its entirety and substitute: "4.5.5 Carrying strap dimensions and strength test. Subject each strap individually to the following tests. Measure the width of each strap. Swing each strap out from the container against its stop. Apply a force equal to 1.5 times the battery's maximum weight at the grasp of each strap. Apply the force in a direction of maximum torque acting on the strap stop. Apply the force continuously for not less than 5 minutes. Examine each strap and the battery for the requirements of 3.8.5 as modified by MIL-PRF-8565/18."

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NOTES

1. Referenced documents. In addition to MIL-PRF-8565, this specification sheet references MIL-PRF-18148/3.

2. Amendment notations. The margins of this specification sheet are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

TABLE I. Qualification inspection and conformance inspection of batteries.

Test number	Examinations and tests	Qualification inspection					Conformance inspection		
		Sample number			Requirement paragraph 1/	Test paragraph 1/	Group A sample 1 100%	Group B sample number	
		1	2	3				1	2
1	INCOMING INSPECTION	X	X	X	3.8.1 (G)	4.5.1 (G)	X	X	X
2	VISUAL AND MECHANICAL EXAMINATION	X	X	X	3.5 (G), 3.6 (G), 3.8.2 (G) table II (G)	4.5.2 (G)	X	X	X
3	DIMENSIONS AND WEIGHT	X	X	X	Reqmt 2 (N), 3.8.3 (G)	4.5.3 (G)	X 2/	X	
4	DIELECTRIC STRENGTH AND INSULATION RESISTANCE TEST	X	X	X	3.7 (G), 3.8.4 (G)	4.5.4 (G)	X 2/	X	X
5	CARRYING STRAP DIMENSIONS AND STRENGTH TEST	X	X	X	3.7 (G), 3.8.5 (S)	4.5.5 (S)			
6	ELECTRICAL TEST PREPARATION	X	X	X	--	4.5.6 (G)		X	X
7	CAPACITY DISCHARGE	X	X	X	Reqmt 3 (N), 3.8.6 (G)	4.5.7 (G)	X	X	X
8	STRENGTH OF RECEPTACLE TEST			X	3.7 (G), 3.8.8 (G)	4.5.9 (G)			
9	CYCLING TEST	X			3.8.9 (G), Reqmt 1 of table III (G)	4.5.10 (G)		X	
10	CAPACITY DISCHARGE AT 49 °C (120 °F)		X		3.8.6 (G), Reqmt 2 of table III (G)	4.5.11 (G)			
11	CHARGE AND DISCHARGE TEST AT LOW TEMPERATURE		X		3.8.6 (G), Reqmt 3 of table III (G)	4.5.12 (G)			X
12	DISCHARGE WHILE INVERTED		X		3.5.9 (G), Reqmt 5 of table III (G)	4.5.13 (G)			X
13	ALTITUDE TEST		X		3.7 (G), 3.5.9 (G), 3.8.10 (G)	4.5.14 (G)			X

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TABLE I. Qualification inspection and conformance inspection of batteries - Continued.

Test number	Examinations and tests	Qualification inspection					Conformance inspection		
		Sample number			Requirement paragraph 1/	Test paragraph 1/	Group A sample 1 100%	Group B sample number	
		1	2	3				1	2
14	SHOCK TEST (BASIC DESIGN)			X	3.7 (G), 3.8.10 (G)	4.5.15 (G)		X	
15	TEMPERATURE SHOCK TEST	X			3.7 (G) 3.8.10 (G)	4.5.16 (G)			
16	TEMPERATURE RISE AND FLOAT TEST		X		3.7 (G), 3.8.11 (G), Reqmt 1 of table III (G)	4.5.17 (G)			X
17	BATTERY GAS EMISSION TEST		X		3.8.10 (G), 3.8.12 (G)	4.5.18 (G)			
18	VIBRATION TEST		X		3.7 (G), 3.8.10 (G)	4.5.19 (G)			
19	HUMIDITY TEST		X		3.7 (G), 3.8.10 (G)	4.5.20 (G)			
20	SALT FOG TEST	X			3.7 (G) 3.8.10 (G)	4.5.21 (G)			
21	GROUND STORAGE TEST			X 3/	3.8.13.1 (G)	4.5.22 (G)			
22	DEEP DISCHARGE RECOVERY TEST			X	3.7 (G), 3.8.13.3 (G)	4.5.27 (G)			
23	PHYSICAL INTEGRITY AT HIGH TEMPERATURE TEST	X			3.7 (G), 3.8.10 (G), Reqmt 1 of table III (G)	4.5.23 (G)			
24	SHOCK TEST (CRASH SAFETY)			X	3.8.10 (G) 3.8.15 (G)	4.5.25 (G)			
25	IMPACT RESISTANCE TEST (NONMETALLIC BATTERY CONTAINER)		4/		3.8.14 (G)	4.5.24 (G)			
26	WORMGEAR RETAINING PIN HARDNESS		5/		3.5.5 (S)	4.5.30 (G)			
27	SHELF LIFE TEST			X	3.8.9 (G), 3.8.13.2 (G)	4.5.26 (G)			
28	FINAL EXAMINATION	X	X	X	3.7 (G), 3.8.4 (G)	4.5.31 (G)		X	X

1/ The notations in parentheses specify the source of the paragraph. “(G)” means the paragraph as shown in MIL-PRF-8565. “(S)” shows that the paragraph has been modified by this specification sheet. “(N)” indicates that a new paragraph or requirement has been added by MIL-PRF-8565/18.

2/ See sample information in test paragraph.

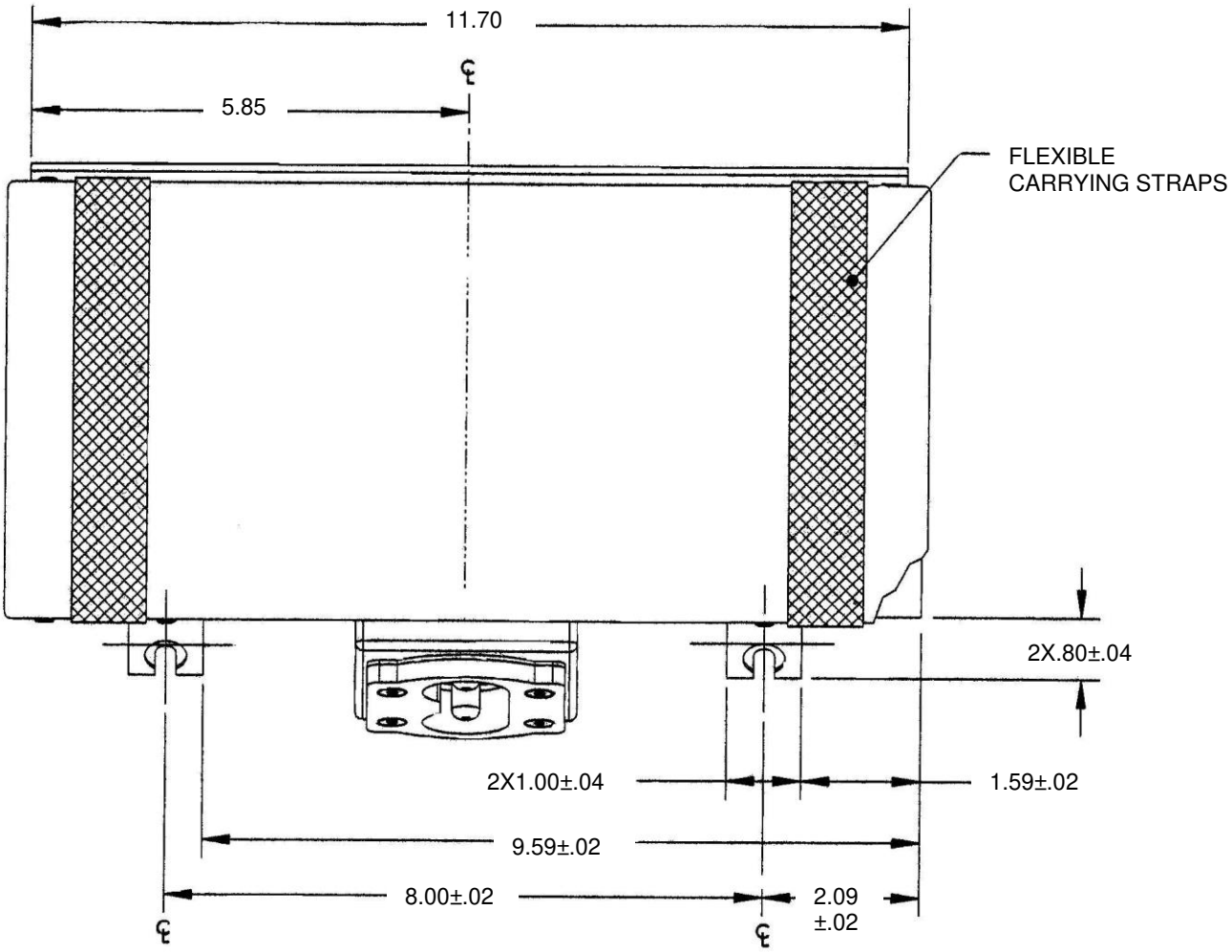
3/ The manufacturer requesting qualification to this specification may submit an additional (fourth) test sample to expedite the qualification test program.

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TABLE I. Qualification inspection and conformance inspection of batteries - Continued.

4/ For nonmetallic battery container designs, two outer nonmetallic battery containers shall be furnished for inspection.

5/ Five wormgear retaining pins shall be furnished for inspection.

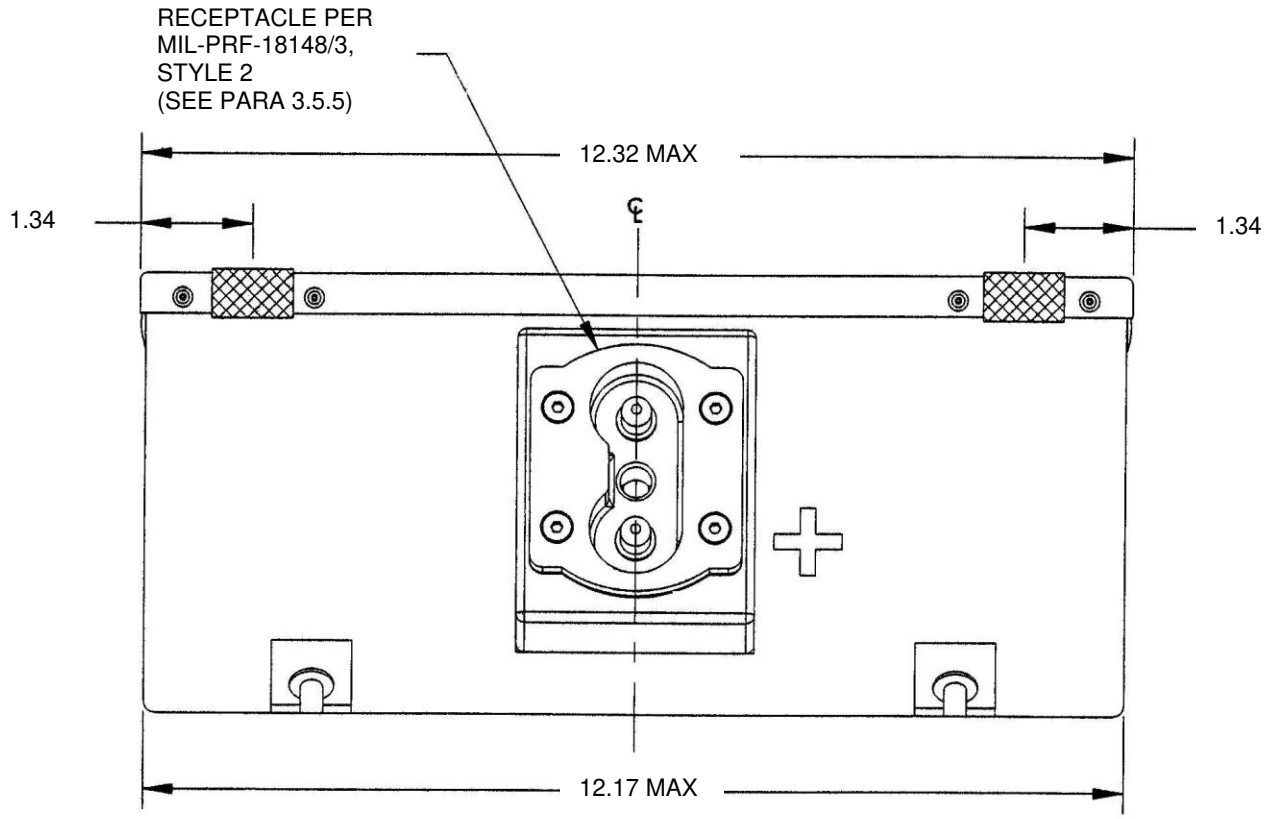


Top View

Notes:
1. All dimensions are in inches.
2. Tolerances: .XX ±.06 inch; .XXX ±.030 inch; angles ±1°.

FIGURE 1. Dimensions and configuration for PIN D8565/18-1 - Continued.

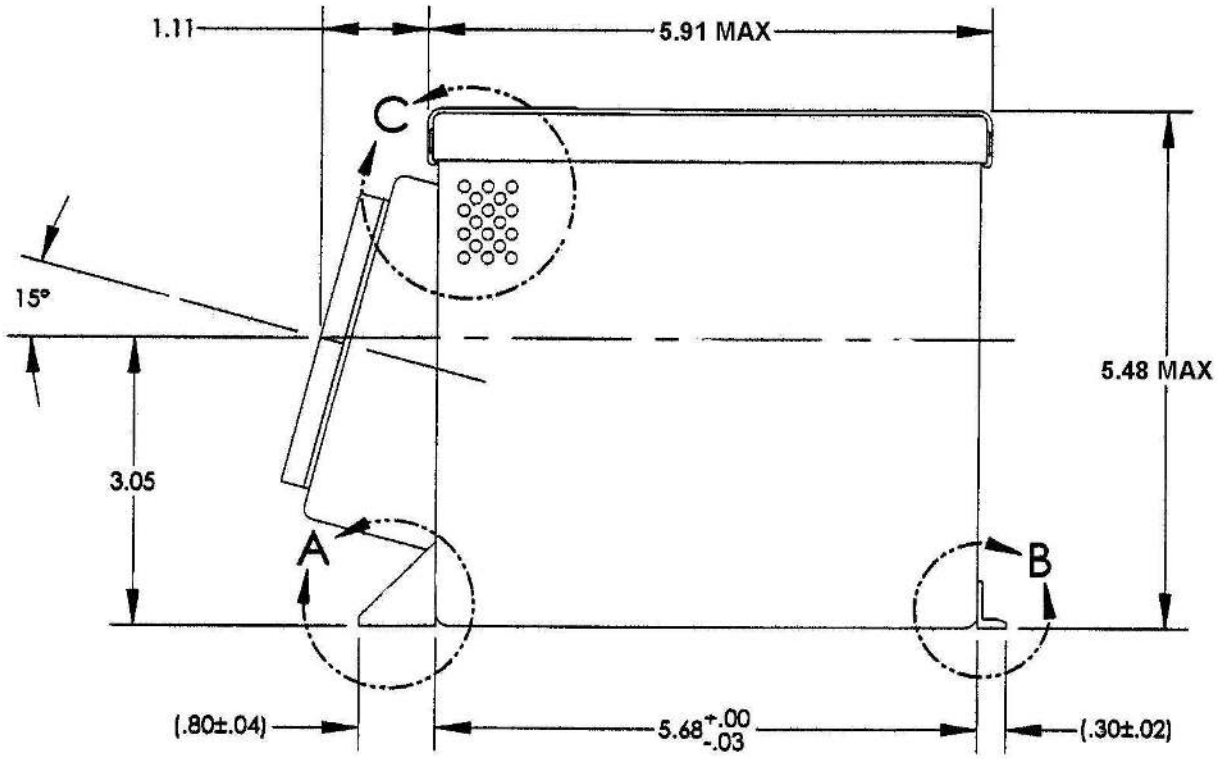
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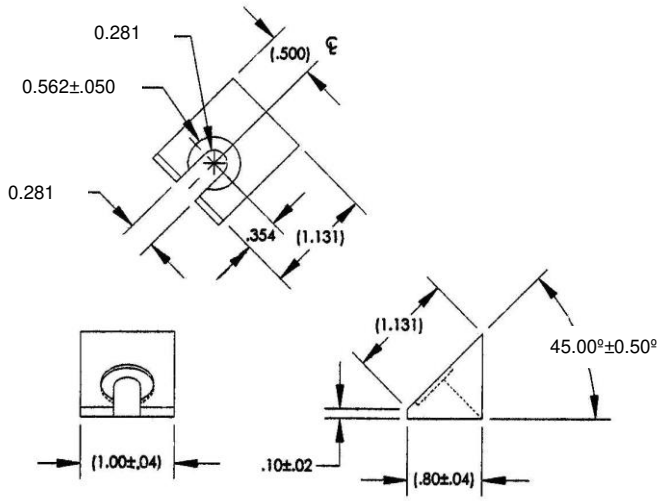
Front View

FIGURE 1. Dimensions and configuration for PIN D8565/18-1 - Continued.

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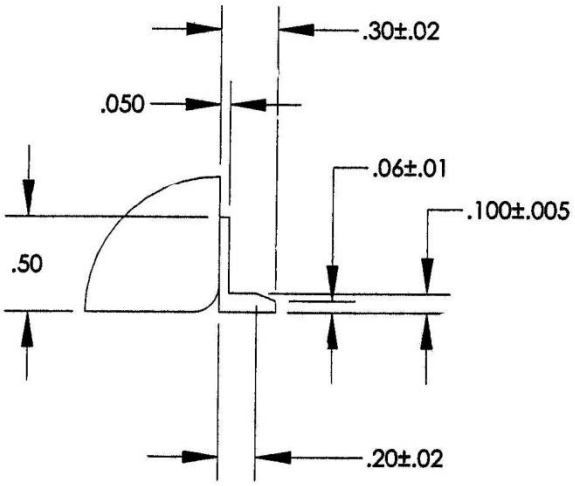
Side View



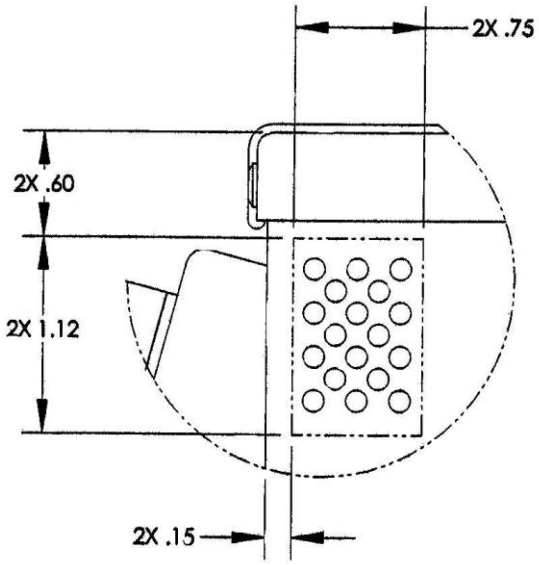
Detail A - Front mounts.

FIGURE 1. Dimensions and configuration for PIN D8565/18-1 - Continued.

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Detail B - Back mounting bar.

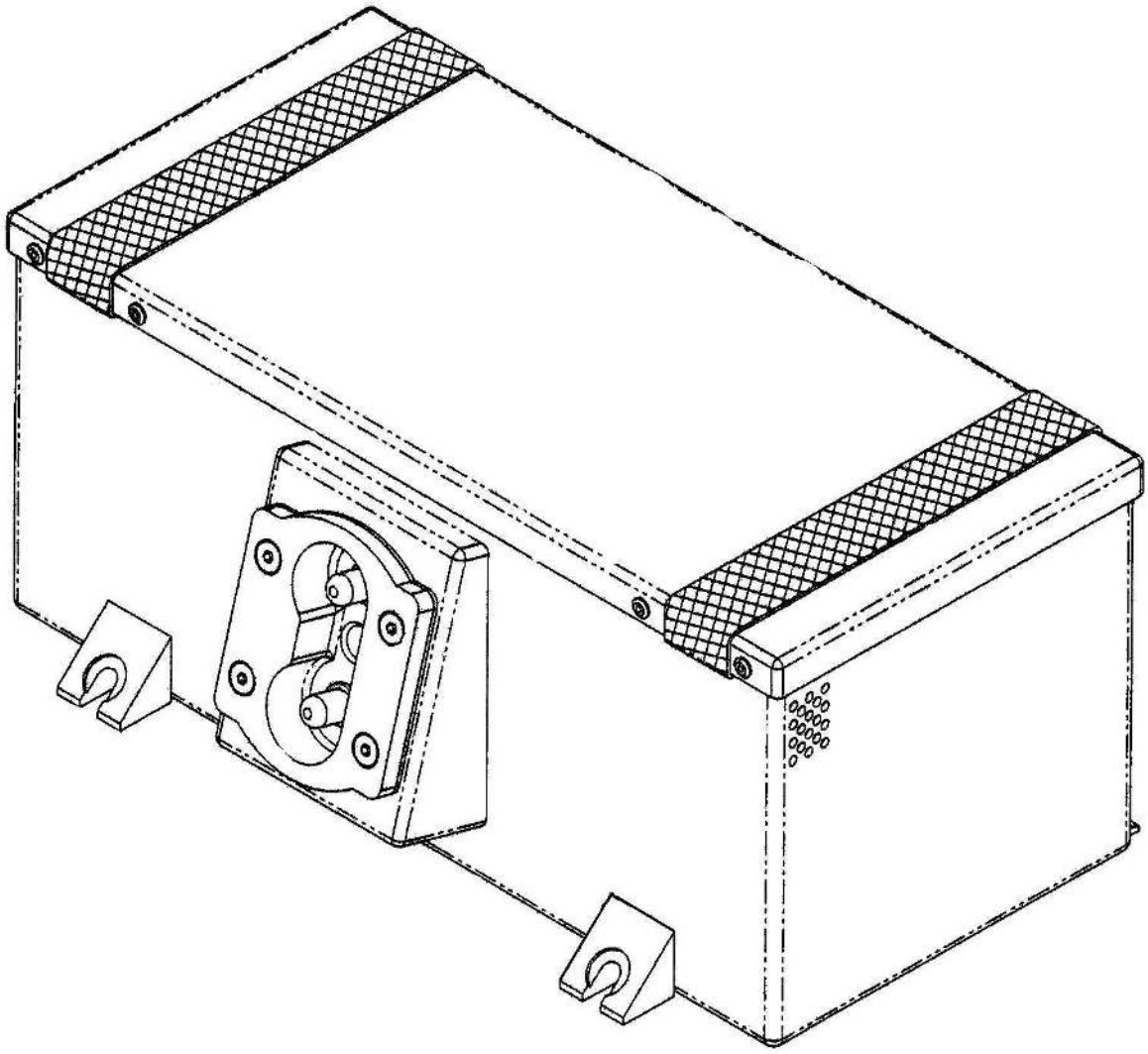


Detail C - Vent holes.

Note: 2X vent hole pattern equally spaced 0.125 max.

FIGURE 1. Dimensions and configuration for PIN D8565/18-1 - Continued.

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Isometric View

FIGURE 1. Dimensions and configuration for PIN D8565/18-1.

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CONCLUDING MATERIAL

Custodians:
Navy - AS
Air Force -170
DLA - CC

Preparing activity:
Navy - AS

(Project 6140-2021-028)

Review activity:
Air Force - 70

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